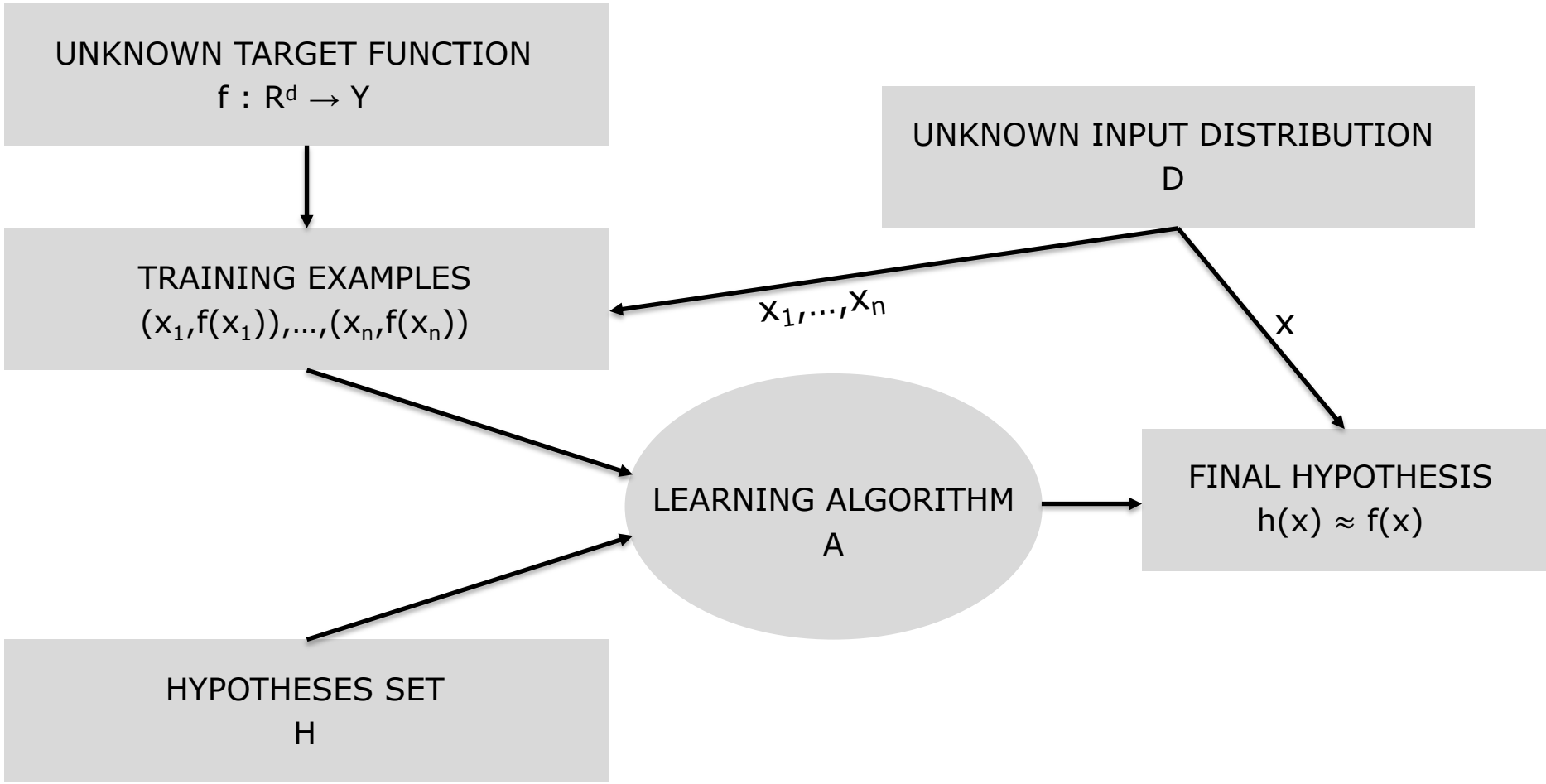
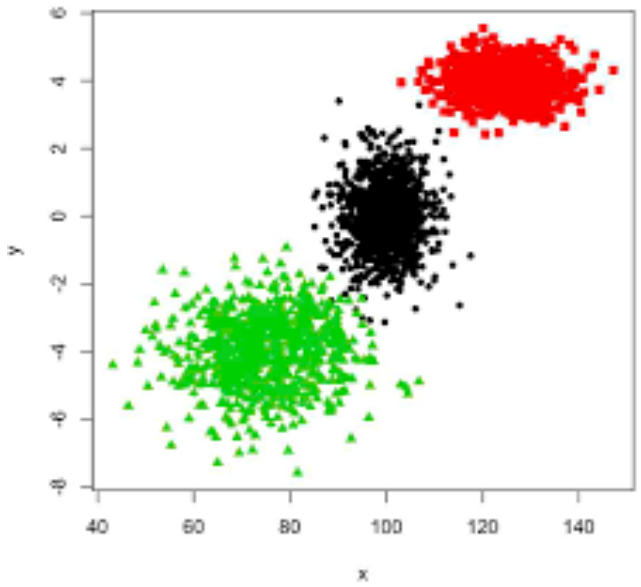
# Principal Component Analysis

# Supervised Learning



# Unsupervised Learning

- Representative problems:
  - Clustering
  - Embeddings




- Clustering: Partition data into similar groups.
- Embeddings: Find good representations of data.

# Difference

- Supervised Learning:
  - Unknown target $f : R^d \rightarrow Y$
  - Input: $(x_1,f(x_1)),\ldots,(x_n,f(x_n))$
  - Goal: Predict f(x) on new x. Learn f.

- Unsupervised Learning:
  - Input: $x_1,\ldots,x_n$
  - Goal:
    - Group/Cluster/Partition based on similarity in $x_i$'s.
    - Alter $x_i$'s to help learning/speed up algorithms.

- Embeddings focus of next lectures.

# Dimensionality Reduction

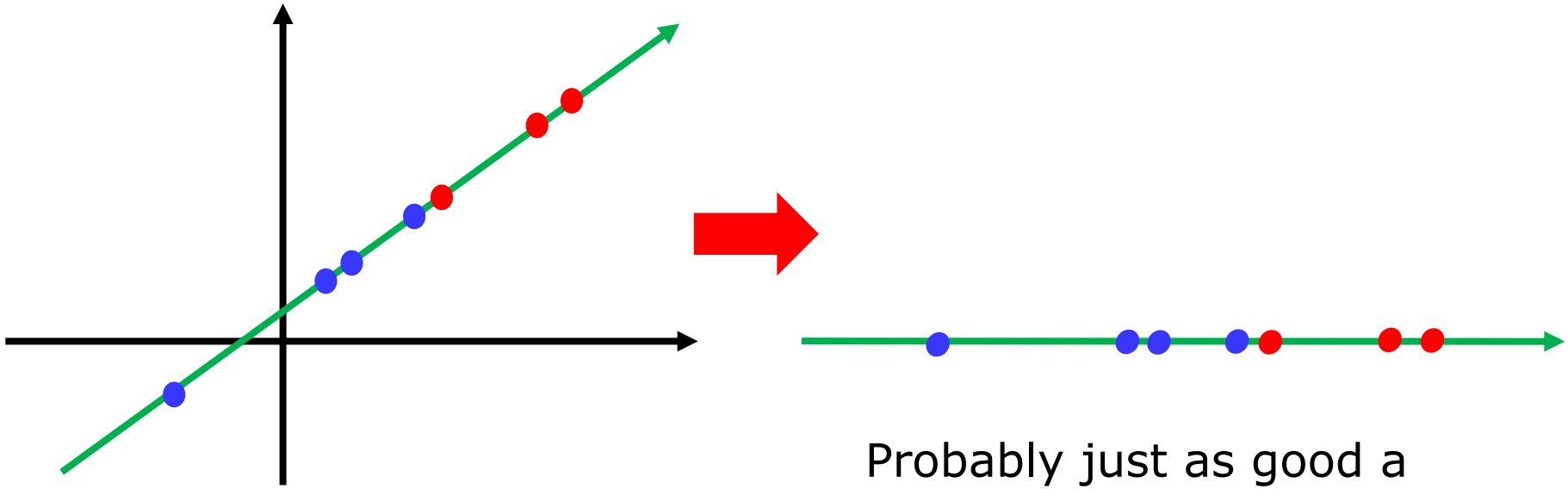
- Redundant Measurements:
  - Data may be redundant:
    - Car: Speed Mph, Fuel, Color, Seats, Speed km/h
  - Or highly correlated:
    - Student: Work Ethics, Grade



- Goal:
  - Remove redundancies.
  - Reduces number of features.
    - Speeds up/helps algorithms.
    - Visualization if very few features (e.g. 2 or 3).
    - Remove noise.

# Redundancy

- 2 features but one-dimensional:



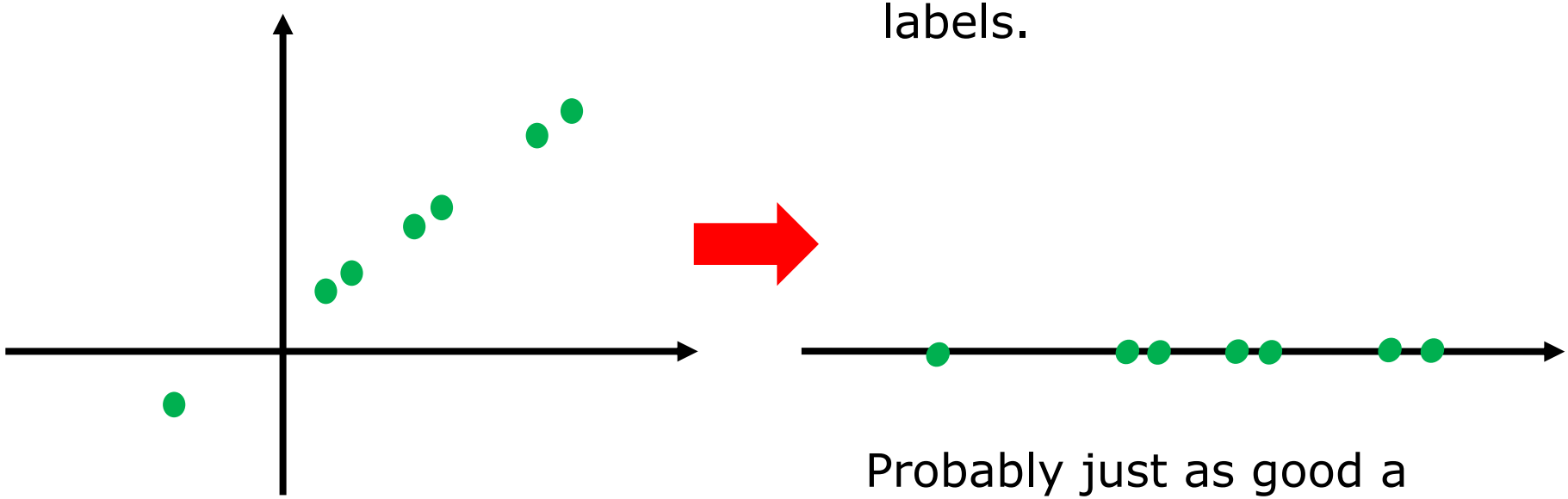
Probably just as good a representation for learning.

# Redundancy

- 2 features but one-dimensional:

Important: Unsupervised, so labels play no role in embedding.

Can be done without knowing labels.



Probably just as good a representation for learning.

# Dimensionality Reduction

- Redundant Measurements:
  - Hand-written digits.
  - 784 pixels = dimensions.
  - No-where near all vectors of 784 dimensions look like digits.

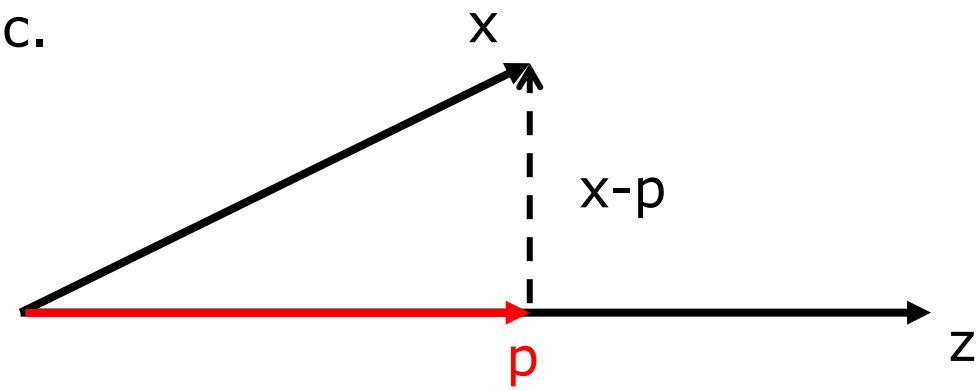


MNIST

- Principal Component Analysis:
  - Project feature vectors *onto k-dimensional subspace*.
  - Only k features needed (write in basis of subspace).

# Linear Algebra: Projections

- Projecting x onto z:
  - x-p orthogonal to z:
    - $(x-p)^T z=0$
  - p = cz for some scaling c.
    - $(x-cz)^T z=0$
    - $x^T z - c z^T z=0$
    - $c = \frac{x^T z}{z^T z}$
  - $p = \frac{x^T z}{z^T z} \cdot z$

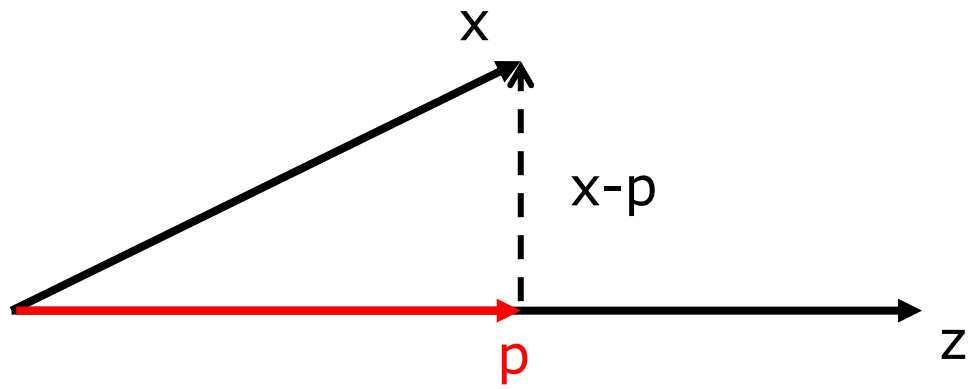



- Simplification:
  - If z has unit norm, then $z^T z=1$ and $p=(x^T z)z$.

Inner product gives length of projection

# Linear Algebra: Projections

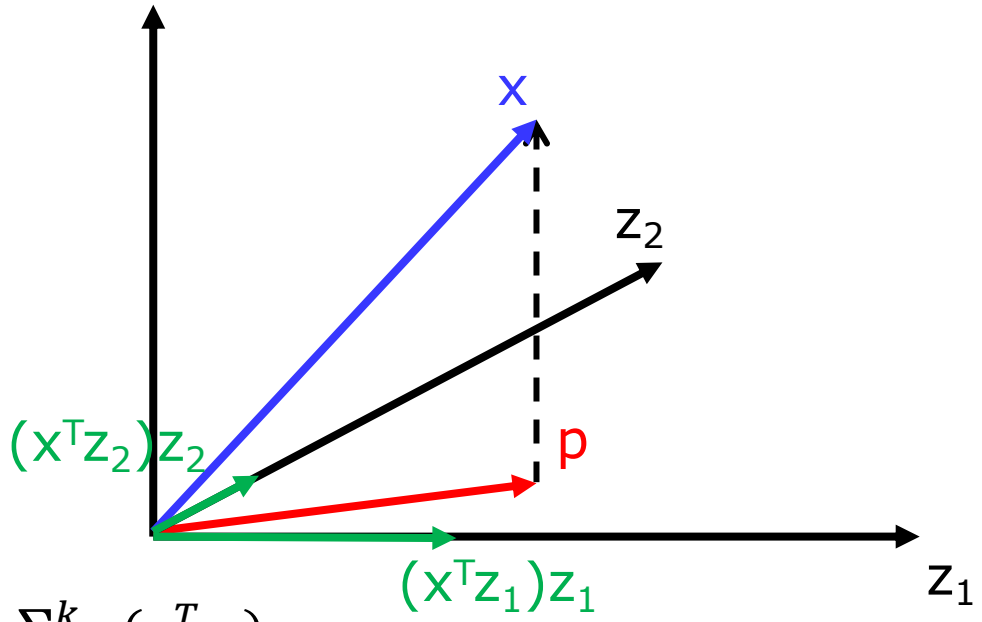
- Projecting x onto unit norm vector z:
  - p=(xTz)z




- Same as:
  - Projecting onto 1-dimensional subspace spanned by z.

# Linear Algebra: Projections

- Projecting x onto k-dimensional subspace:
  - Let $z_1,\ldots,z_k$ be orthonormal basis for subspace.
    - $z_i^T z_j = 0$ for $i \neq j$
    - $z_i^T z_i = 1$ for all i (unit length)




  - $p = \sum_{i=1}^{k} (x^T z_i) z_i$

# PCA

- Principal Component Analysis:
  - Project feature vectors *onto k-dimensional subspace*.
  - Equivalently, find k orthogonal unit vectors $z_1,\ldots,z_k$ and project feature vector x to $p = \sum_{i=1}^{k} (x^T z_i) z_i$.
  - Write x in the new basis:
    - $\hat{x} = \begin{pmatrix} x^T z_1 \\ \vdots \\ x^T z_k \end{pmatrix}$
  - Reduced #features to k.
- Question:
  - What basis $z_1,\ldots,z_k$ to choose?



MNIST



# Principal Component Analysis

# Dimensionality Reduction

- Redundant Measurements:
  - Data may be redundant:
    - Car: Speed Mph, Fuel, Color, Seats, Speed km/h
  - Or highly correlated:
    - Student: Work Ethics, Grade



- Goal:
  - Remove redundancies.
  - Reduces number of features.
    - Speeds up/helps algorithms.
    - Visualization if very few features (e.g. 2 or 3).

# PCA

- Principal Component Analysis:
  - Project feature vectors *onto k-dimensional subspace*.
  - Equivalently, find k orthogonal unit vectors $z_1,\ldots,z_k$ and project feature vector x to $p = \sum_{i=1}^{k} (x^T z_i) z_i$.
  - Write x in the new basis:
    - $\hat{x} = \begin{pmatrix} x^T z_1 \\ \vdots \\ x^T z_k \end{pmatrix}$
  - Reduced #features to k.
- Question:
  - What basis $z_1,\ldots,z_k$ to choose?

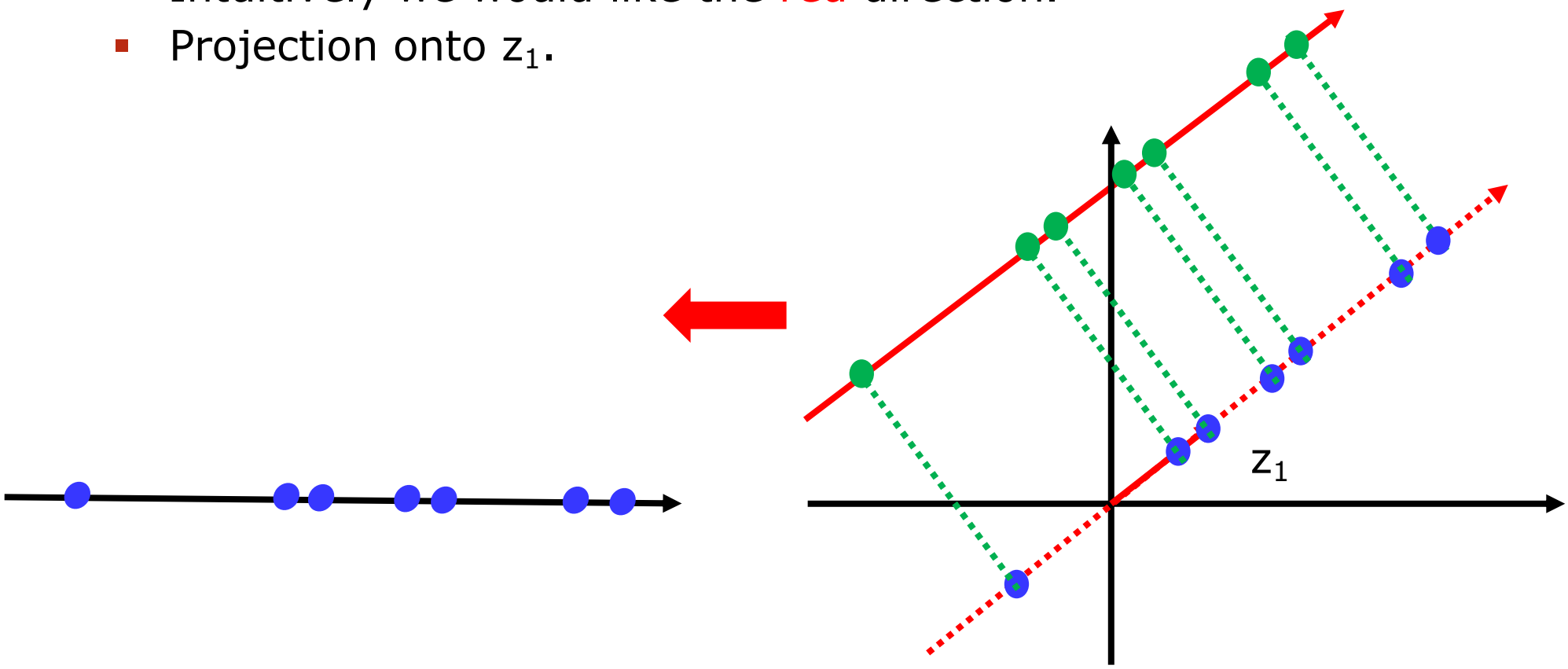


MNIST



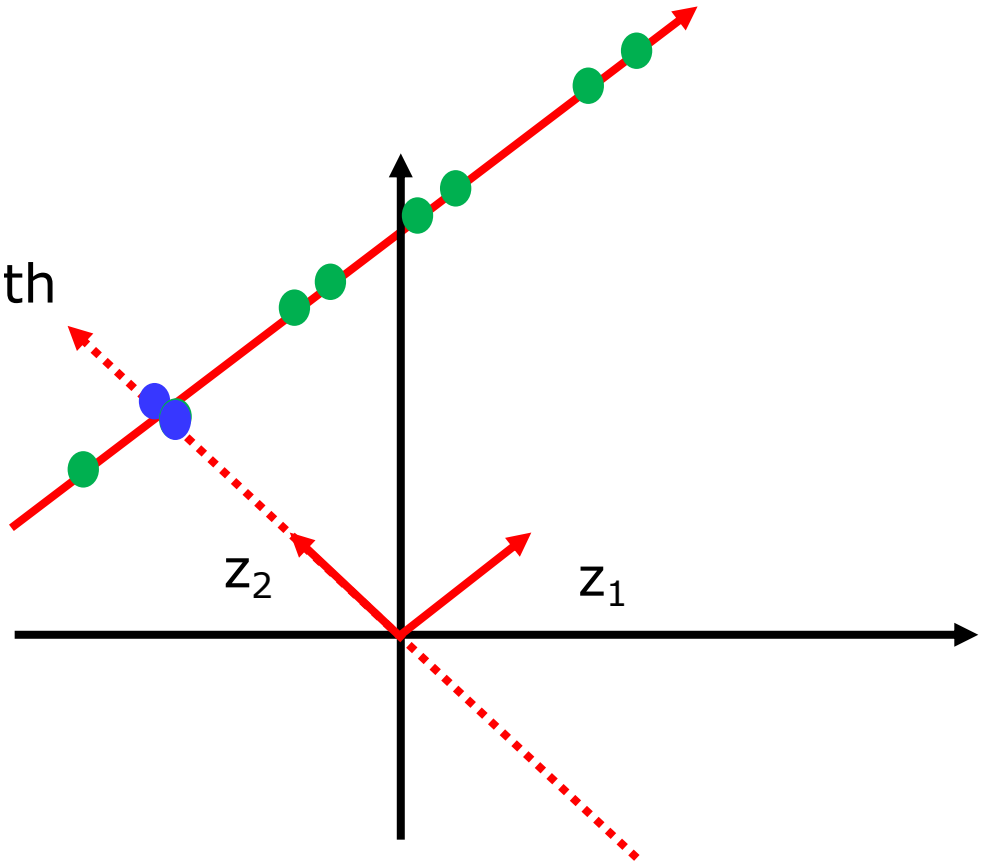
# Choice of Basis

- Motivating Example:
  - Intuitively we would like the red direction.
  - Projection onto $z_1$.



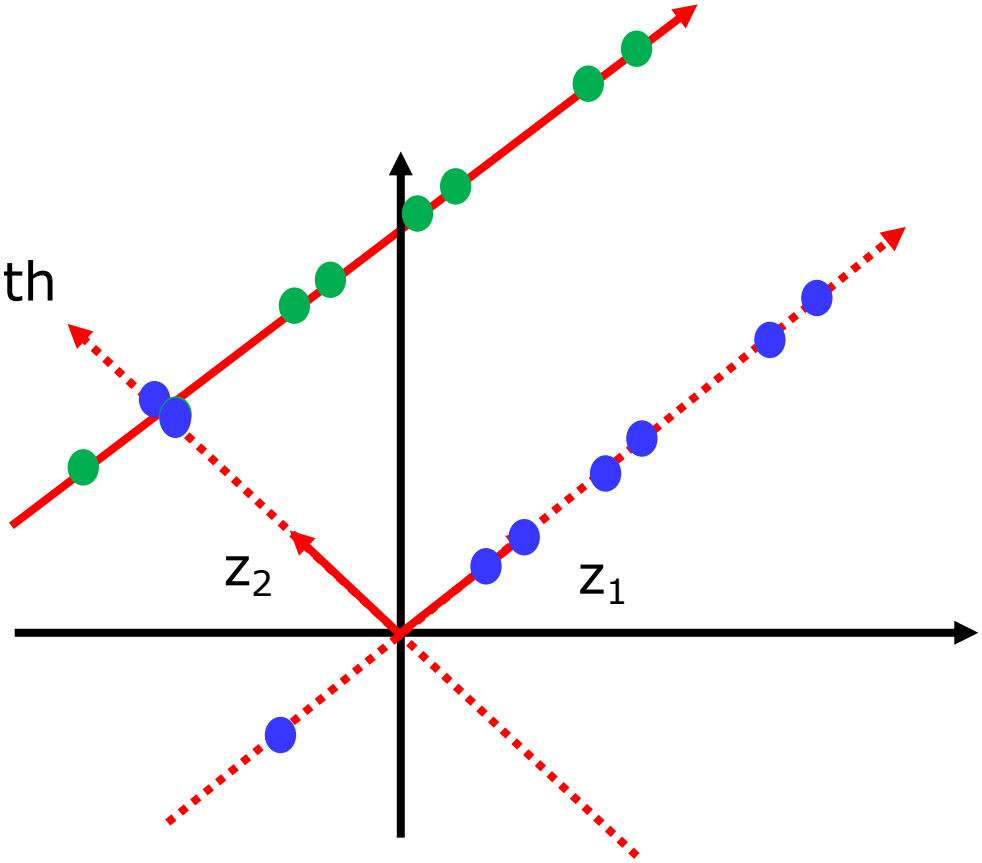
# Choice of Basis

- Motivating Example:
  - Intuitively we would like the red direction.
  - Projection onto $z_1$.
- What formally characterizes $z_1$?
  - Why better than $z_2$?
  - Points have large norms in both projections.



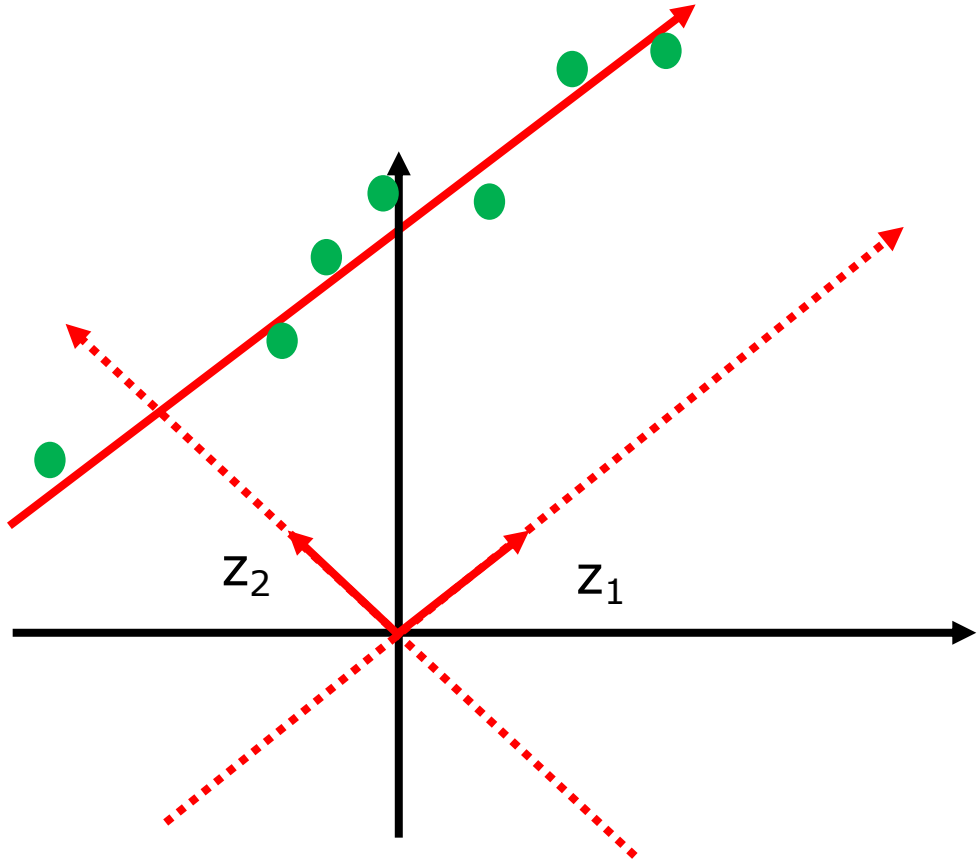
# Choice of Basis

- Motivating Example:
  - Intuitively we would like the red direction.
  - Projection onto $z_1$.
- What formally characterizes $z_1$?
  - Why better than $z_2$?
  - Points have large norms in both projections.
  - But more "spread out" for $z_1$



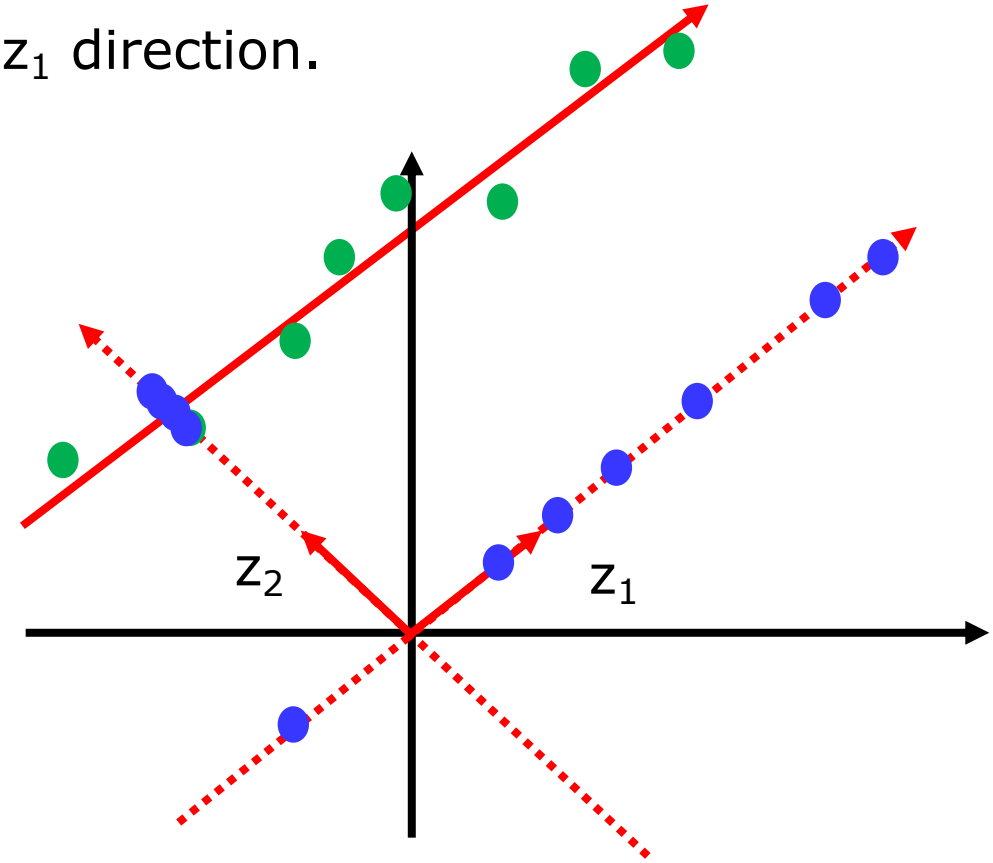
# Choice of Basis

- Motivating Example:
  - What if a bit of noise in data?



# Choice of Basis

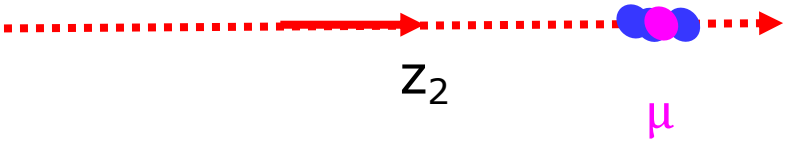
- Motivating Example:
  - What if a bit of noise in data?
    - Still most ”spread out” in $z_1$ direction.

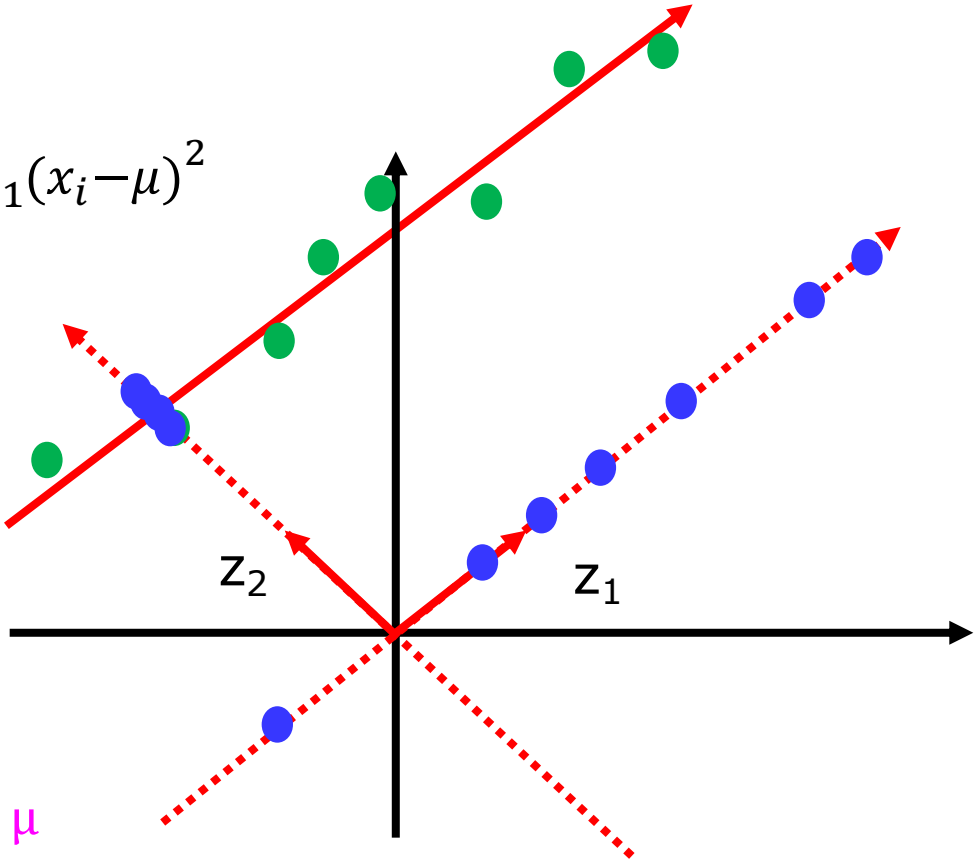


# Choice of Basis

- Defining "Spread Out":
  - For real values $x_1,\ldots,x_n$:
    - Sample mean $\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$
    - Sample variance $\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i-\mu)^2$

- Small sample variance in projections onto $z_2$:



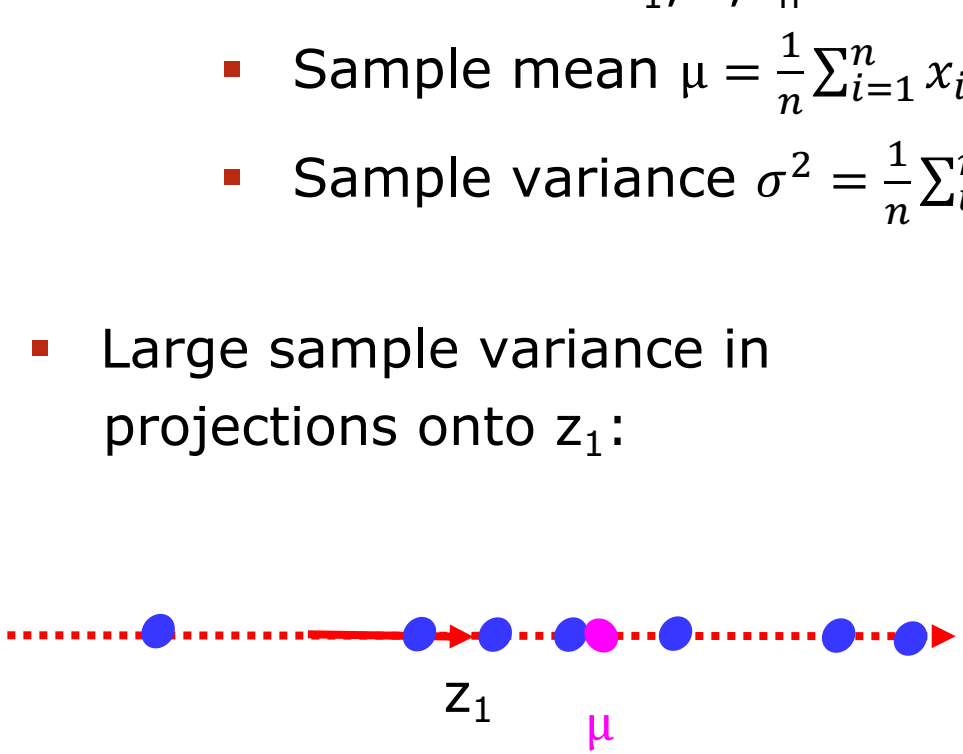



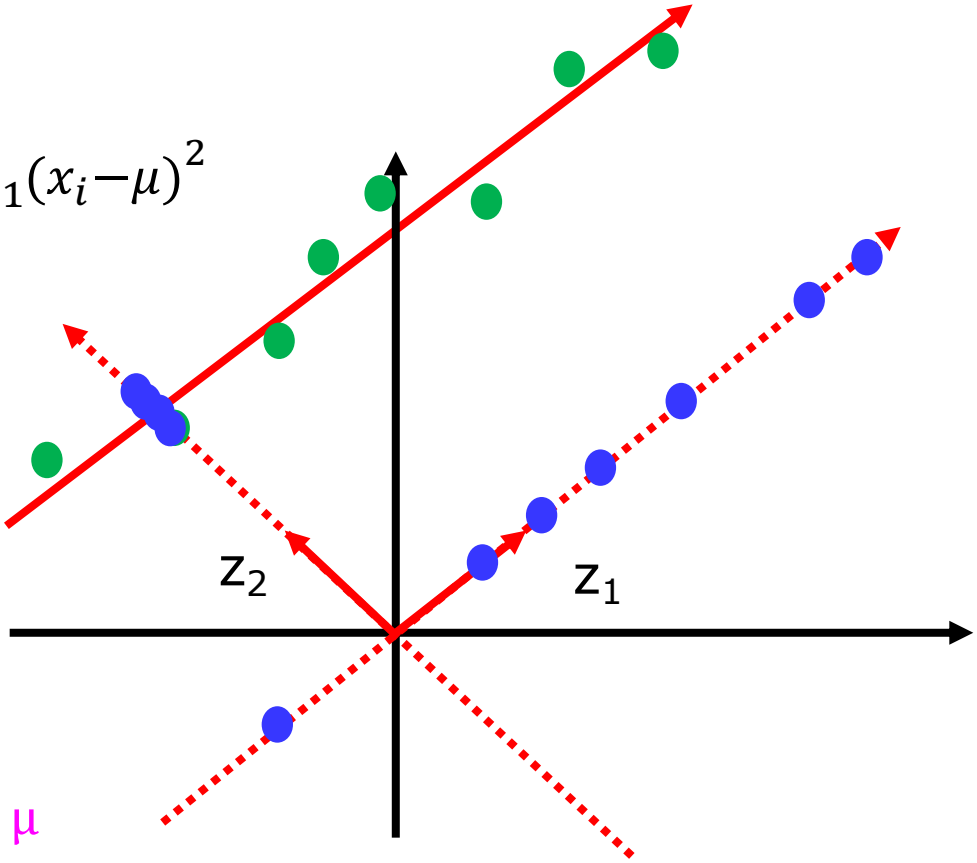

$\sigma^2$ is average squared distances from $\mu$

# Choice of Basis

- Defining "Spread Out":
  - For real values $x_1,\ldots,x_n$:
    - Sample mean $\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$
    - Sample variance $\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i-\mu)^2$

- Large sample variance in projections onto $z_1$:

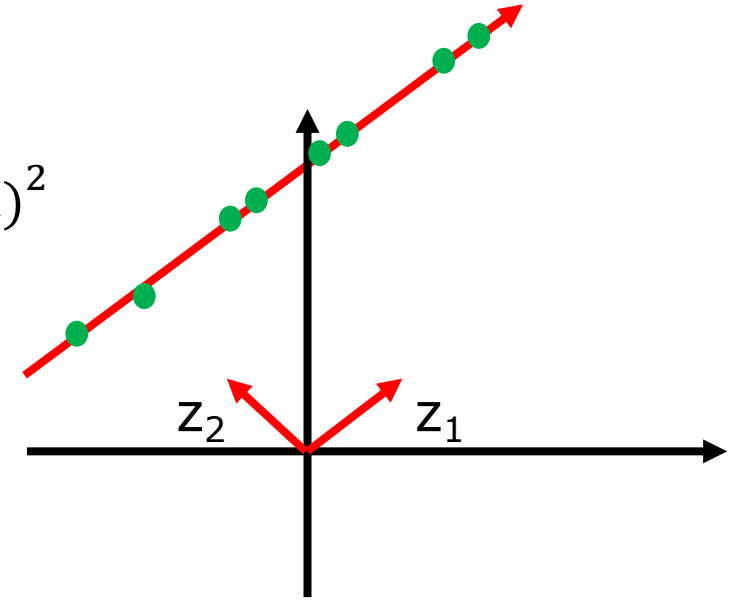



$\sigma^2$ is average squared distances from $\mu$

# Choice of Basis

- Defining "Spread Out":
  - For real values $x_1,\ldots,x_n$:
    - Sample mean $\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$
    - Sample variance $\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i-\mu)^2$




- Hope:
  - Direction with largest variance has most information about data.
  - Sometimes referred to as the "signal-to-noise" ratio is largest.

# PCA

- Principal Component Analysis:
  - Project feature vectors *onto k-dimensional subspace*.
  - Equivalently, find k orthogonal unit vectors $z_1,\ldots,z_k$ and project feature vector x to $p = \sum_{i=1}^{k} (x^T z_i) z_i$.
  - Write x in the new basis:
    - $\hat{x} = \begin{pmatrix} x^T z_1 \\ \vdots \\ x^T z_k \end{pmatrix}$
  - Reduced #features to k.
- Question:
  - What basis $z_1,\ldots,z_k$ to choose?
    - Maximize variance!

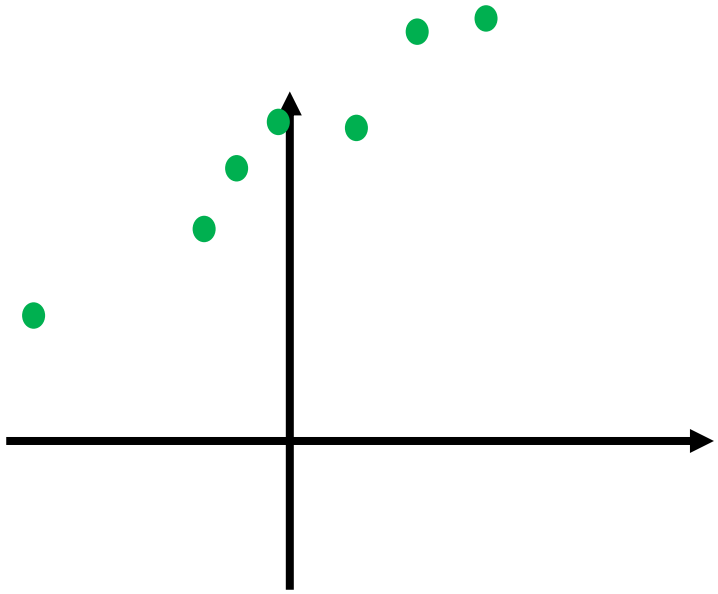


MNIST

# PCA

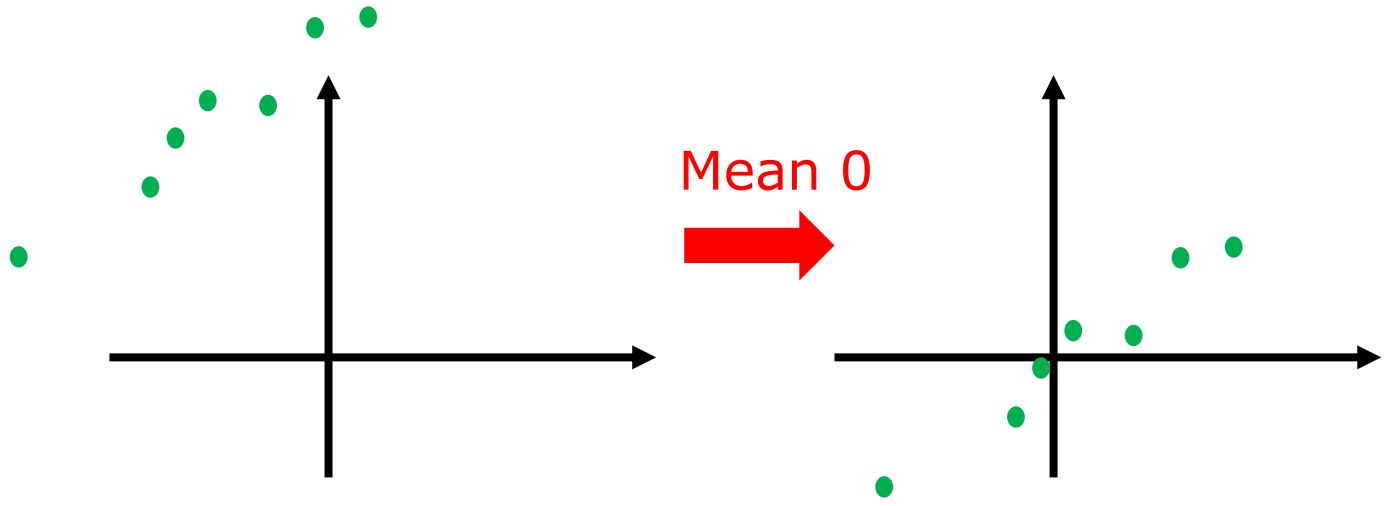
- PCA: Choosing k directions $z_1,\ldots z_k$ for projection greedily:
  - For i=1,…,k:
    - Let $z_i$ be unit length such that:
      - $z_i$ is orthogonal to $z_1,\ldots,z_{i-1}$.
      - Sample variance of $x_1^T z_i,\ldots,x_n^T z_i$ is largest among all unit vectors z that are orthogonal to $z_1,\ldots,z_{i-1}$.

- Question:
  - How can we compute $z_i$ efficiently??
    - Let us simplify a bit first.



# Simplification by Mean Zero

- Simplifying Step:
  - Start by making the mean of every feature 0.

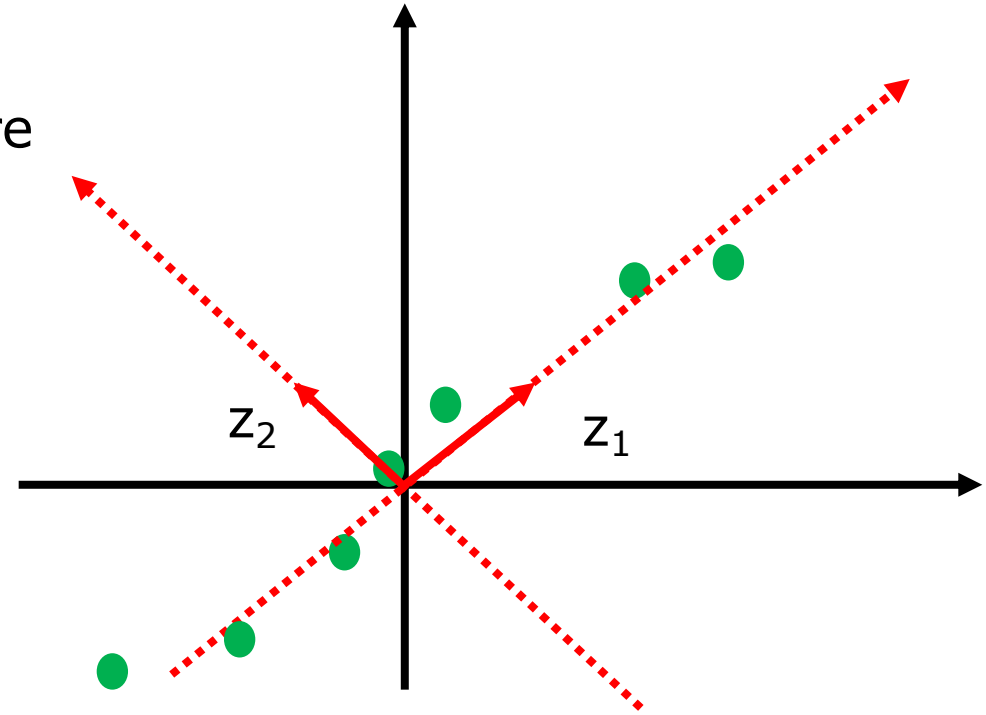



  - $\begin{pmatrix} x_{1,1} & \cdots & x_{1,j} & \cdots & x_{1,d} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ x_{n,1} & \cdots & x_{n,j} & \cdots & x_{n,d} \end{pmatrix}$
  - Subtract $\sum_{i=1}^{n} \frac{x_{i,j}}{n}$ from each entry in j'th column.

# Simplification by Mean Zero

- Defining "Spread Out" in direction $z_j$:
  - Sample mean $\mu = \frac{1}{n}\sum_{i=1}^{n} x_i^T z_j$
  - Sample variance $\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i^T z_j - \mu)^2$

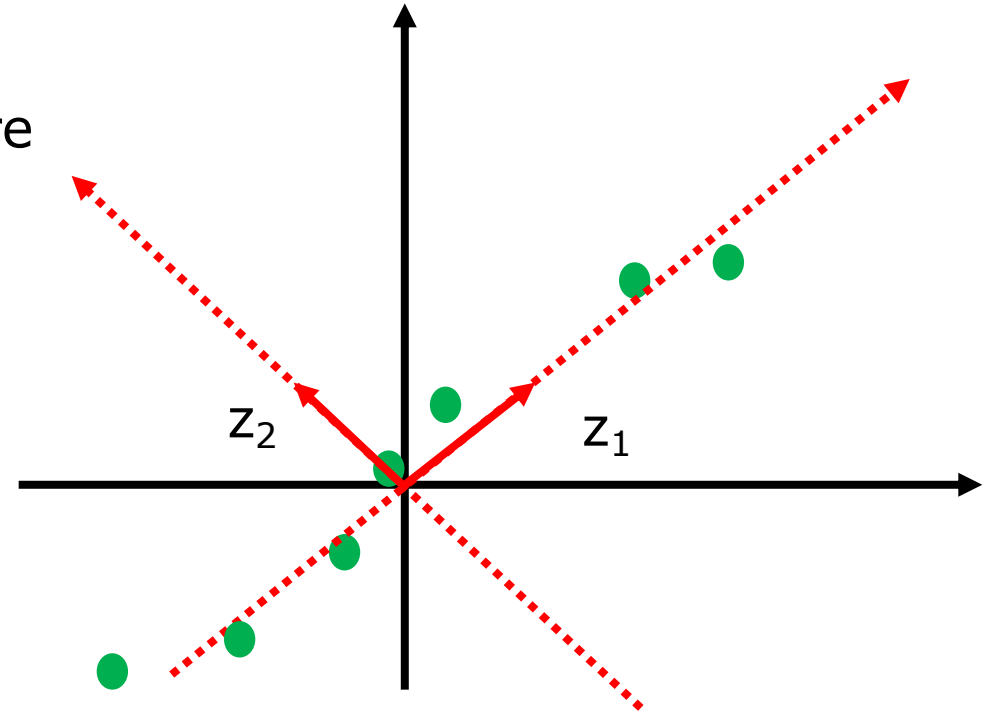
- After setting mean of each feature to 0:
- $\mu = \frac{1}{n}\sum_{i=1}^{n} x_i^T z_j$
- $= \frac{1}{n}\sum_{i=1}^{n}\sum_{k=1}^{d} x_{i,k} z_{j,k}$
- $= \frac{1}{n}\sum_{k=1}^{d} z_{j,k} \sum_{i=1}^{n} x_{i,k}$
- $= 0$



# Simplification by Mean Zero

- Defining "Spread Out" in direction $z_j$:
  - Sample mean $\mu = \frac{1}{n}\sum_{i=1}^{n} x_i^T z_j$
  - Sample variance $\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i^T z_j - \mu)^2$

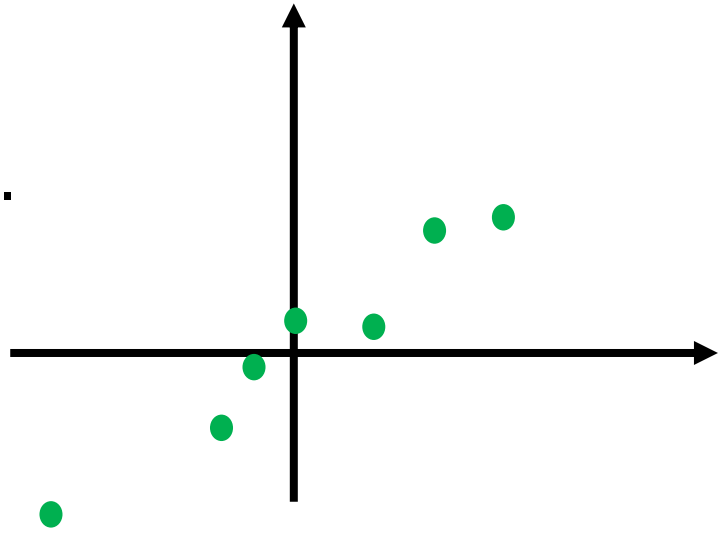
- After setting mean of each feature to 0:
- $\mu = 0$
- $\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i^T z_j)^2$



# PCA

- PCA: Choosing k directions $z_1,\ldots z_k$ for projection greedily:
  - For i=1,…,k:
    - Let $z_i$ be unit length such that:
      - $z_i$ is orthogonal to $z_1,\ldots,z_{i-1}$.
      - Sample variance of $x_1^T z_i,\ldots,x_n^T z_i$ is largest among all unit vectors z that are orthogonal to $z_1,\ldots,z_{i-1}$.

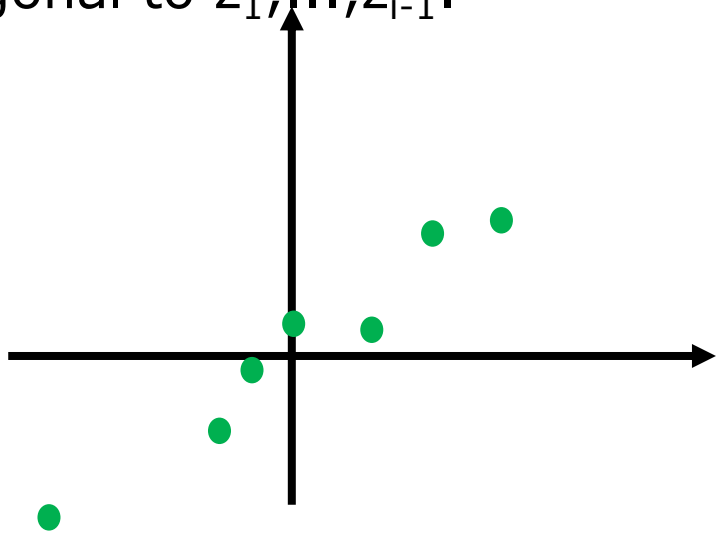
- Question:
  - How can we compute $z_i$ efficiently??
    - First make mean 0 for each feature.



# PCA

- PCA: Choosing k directions $z_1,\ldots z_k$ for projection greedily:
  - Make mean of each feature 0.
  - For i=1,…,k:
    - Let $z_i$ be unit length such that:
      - $z_i$ is orthogonal to $z_1,\ldots,z_{i-1}$.
      - Sample variance $\frac{1}{n}\sum_{j=1}^{n}(x_j^T z_i)^2$ is largest among all unit vectors z that are orthogonal to $z_1,\ldots,z_{i-1}$.
- Linear Algebra Fact:

Let $X = \begin{pmatrix} - & x_1 & - \\ \vdots & \vdots & \vdots \\ - & x_n & - \end{pmatrix}$ then $z_i$ equals the eigenvector $\mu_i$ corresponding to the i'th largest eigenvalue of $X^TX$.



# Eigenvalues and Eigenvectors

- Linear Algebra Reminder:
  - Let A be n x n matrix.
  - Then $\mu$ is an eigenvector with eigenvalue $\lambda$ if
    - A $\mu = \lambda\mu$.

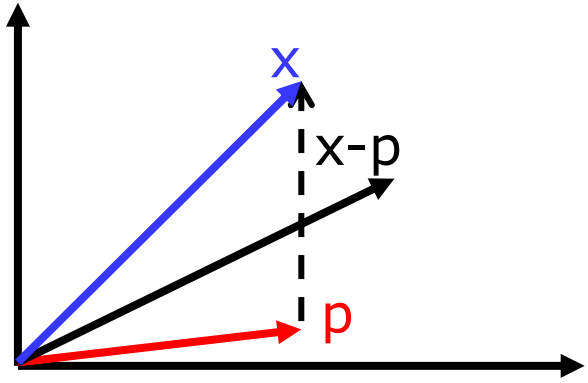
- There are efficient algorithms for computing eigenvectors and eigenvalues.

# PCA

- Principal Component Analysis
  - Let $z_1,\ldots,z_k$ be k eigenvectors corresponding to largest eigenvalues of $X^TX$ and project feature vector x to $p_x = \sum_{i=1}^{k}(x^T z_i)z_i$.
- Write x in the new basis:
  - $\hat{x} = \begin{pmatrix} x^T z_1 \\ \vdots \\ x^T z_k \end{pmatrix}$



MNIST




- Next Up: We will derive PCA via a different motivation and also prove why those $z_1,\ldots,z_k$ are "best".

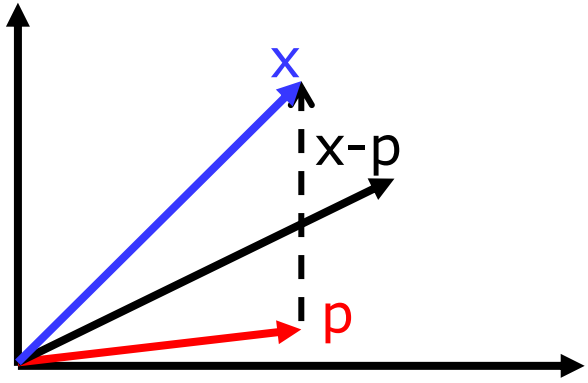
# Principal Component Analysis

# PCA

- Principal Component Analysis
  - Let $z_1,\ldots,z_k$ be k eigenvectors corresponding to largest eigenvalues of $X^TX$ and project feature vector x to $p_x = \sum_{i=1}^{k}(x^T z_i)z_i$.
- Write x in the new basis:
  - $\hat{x} = \begin{pmatrix} x^T z_1 \\ \vdots \\ x^T z_k \end{pmatrix}$



MNIST




- Next Up: We will derive PCA via a different motivation and also argue why those $z_1,\ldots,z_k$ are "best".

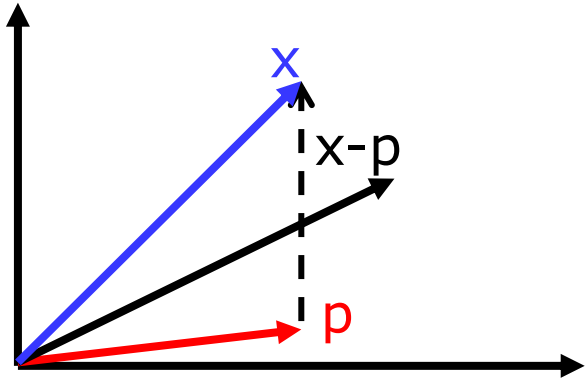
# PCA

- Basis?
  - Find k orthogonal unit vectors $z_1,\ldots,z_k$ and project feature vector x to $p_x = \sum_{i=1}^{k}(x^T z_i)z_i$.



MNIST

- Question:
  - What basis $z_1,\ldots,z_k$ to choose?
    - Maximizing variance one choice.
- Another Motivation for PCA:
  - We lose information when projecting.
    - $x-p_x$ is ”lost”.
  - PCA: Given data $x_1,\ldots,x_n$ minimize
    - $\sum_{j=1}^{n} \left\| x_j - p_{x_j} \right\|^2$ Kind of least squares loss



# PCA

- PCA: Given data $x_1,\dots,x_n$ choose $z_1,\dots,z_k$ to minimize
  - $\sum_{j=1}^{n}\left\|x_j-\sum_{i=1}^{k}(x_j^T z_i)z_i\right\|^2$
- Question:
  - How can we compute such $z_1,\dots,z_k$?
- Minimize:
  - $\sum_{j=1}^{n}\left\|x_j-\sum_{i=1}^{k}(x_j^T z_i)z_i\right\|^2 =$
  - $\sum_{j=1}^{n}\left(x_j-\sum_{i=1}^{k}(x_j^T z_i)z_i\right)^T\left(x_j-\sum_{i=1}^{k}(x_j^T z_i)z_i\right) =$
  - $\sum_{j=1}^{n}(x_j^T x_j-2\sum_{i=1}^{k}(x_j^T z_i)(x_j^T z_i)+\sum_{l=1}^{k}\sum_{i=1}^{k}\left((x_j^T z_i)z_i\right)^T(x_j^T z_l)z_l)$

Does not depend on z's

z's orthogonal, so cross terms disappear.

# PCA

- PCA: Given data $x_1,\dots,x_n$ choose $z_1,\dots,z_k$ to minimize
  - $\sum_{j=1}^{n} \left\| x_j - \sum_{i=1}^{k} (x_j^T z_i) z_i \right\|^2$

- Question:
  - How can we compute such $z_1,\dots,z_k$?
- Minimize:
  - $\sum_{j=1}^{n} (-2 \sum_{i=1}^{k} (x_j^T z_i)^2 + \sum_{i=1}^{k} (x_j^T z_i)^2 z_i^T z_i) =$
  - $\sum_{j=1}^{n} (-2 \sum_{i=1}^{k} (x_j^T z_i)^2 + \sum_{i=1}^{k} (x_j^T z_i)^2) =$
  - $-\sum_{j=1}^{n} \sum_{i=1}^{k} (x_j^T z_i)^2$
- Same as maximizing:
  - $\sum_{j=1}^{n} \sum_{i=1}^{k} (x_j^T z_i)^2$

z's have unit norm

Maximizing sum of sample variances if mean of features 0!

# PCA

- PCA: Given data $x_1,\dots,x_n$ choose $z_1,\dots,z_k$ to minimize
  - $\sum_{j=1}^{n} \left\| x_j - \sum_{i=1}^{k} (x_j^T z_i) z_i \right\|^2$

- Question:
  - How can we compute such $z_1,\dots,z_k$?
- Maximize:
  - $\sum_{j=1}^{n} \sum_{i=1}^{k} (x_j^T z_i)^2$

Square of entry (j,i) in XZ

- Matrix Notation:
  - Let $X = \begin{pmatrix} - & x_1 & - \\ \vdots & \vdots & \vdots \\ - & x_n & - \end{pmatrix}$, $Z = \begin{pmatrix} | & \dots & | \\ z_1 & \dots & z_k \\ | & \dots & | \end{pmatrix}$

# PCA

- PCA: Given data $x_1,\dots,x_n$ choose $z_1,\dots,z_k$ to minimize
  - $\sum_{j=1}^{n} \left\| x_j - \sum_{i=1}^{k} (x_j^T z_i) z_i \right\|^2$
- Question:
  - How can we compute such $z_1,\dots,z_k$?
- Maximize:
  - $\sum_{j=1}^{n} \sum_{i=1}^{k} (x_j^T z_i)^2 = \sum_{j=1}^{n} \sum_{i=1}^{k} (XZ)_{j,i}^2$
- Matrix Notation:
  - Let $X = \begin{pmatrix} - & x_1 & - \\ \vdots & \vdots & \vdots \\ - & x_n & - \end{pmatrix}$, $Z = \begin{pmatrix} | & \dots & | \\ z_1 & \dots & z_k \\ | & \dots & | \end{pmatrix}$

For mxn matrix A, i'th diagonal entry of $A^TA$ equals $\sum_{j=1}^{m} a_{j,i}^2$ i.e. sum of squared entries in i'th column of A.

# PCA

- $A = \begin{pmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,n} \end{pmatrix}$

- $A^T A = \begin{pmatrix} a_{1,1} & \cdots & a_{m,1} \\ \vdots & \cdots & \vdots \\ a_{1,i} & \cdots & a_{m,i} \\ \vdots & \cdots & \vdots \\ a_{1,n} & \cdots & a_{m,n} \end{pmatrix} \begin{pmatrix} a_{1,1} & \cdots & a_{1,i} & \cdots & a_{1,n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{m,1} & \cdots & a_{m,i} & \cdots & a_{m,n} \end{pmatrix}$

- $(A^T A)_{i,i} = \sum_{j=1}^{m} a_{j,i}\, a_{j,i} = \sum_{j=1}^{m} a_{j,i}^2$

# PCA

- PCA: Given data $x_1,\dots,x_n$ choose $z_1,\dots,z_k$ to minimize
  - $\sum_{j=1}^{n} \left\| x_j - \sum_{i=1}^{k} (x_j^T z_i) z_i \right\|^2$
- Question:
  - How can we compute such $z_1,\dots,z_k$?
- Maximize:
  - $\sum_{j=1}^{n} \sum_{i=1}^{k} (x_j^T z_i)^2 = \sum_{j=1}^{n} \sum_{i=1}^{k} (XZ)_{j,i}^2$
- Matrix Notation:
  - Let $X = \begin{pmatrix} - & x_1 & - \\ \vdots & \vdots & \vdots \\ - & x_n & - \end{pmatrix}$, $Z = \begin{pmatrix} | & \dots & | \\ z_1 & \dots & z_k \\ | & \dots & | \end{pmatrix}$

For mxn matrix A, i'th diagonal entry of $A^TA$ equals $\sum_{j=1}^{m} a_{j,i}^2$ i.e. sum of squared entries in i'th column of A.

tr($A^TA$) is sum of diagonals and thus sum of all squared entries in A.

# PCA

- PCA: Given data $x_1,\dots,x_n$ choose $z_1,\dots,z_k$ to minimize
  - $\sum_{j=1}^{n} \left\| x_j - \sum_{i=1}^{k} (x_j^T z_i) z_i \right\|^2$
- Question:
  - How can we compute such $z_1,\dots,z_k$?
- Maximize:
  - $\sum_{j=1}^{n} \sum_{i=1}^{k} (x_j^T z_i)^2 = \sum_{j=1}^{n} \sum_{i=1}^{k} (XZ)_{j,i}^{\ 2} = tr\left((XZ)^T XZ\right) = tr(Z^T X^T XZ)$
- Matrix Notation:
  - Let $X = \begin{pmatrix} - & x_1 & - \\ \vdots & \vdots & \vdots \\ - & x_n & - \end{pmatrix}$, $Z = \begin{pmatrix} | & \dots & | \\ z_1 & \dots & z_k \\ | & \dots & | \end{pmatrix}$

# PCA

- PCA: Given data $x_1,\dots,x_n$ choose $z_1,\dots,z_k$ to minimize
  - $\sum_{j=1}^{n} \left\| x_j - \sum_{i=1}^{k} (x_j^T z_i) z_i \right\|^2$

- Question:
  - How can we compute such $z_1,\dots,z_k$?
- Maximize:
  - $tr(Z^T X^T X Z)$

- Matrix Notation:
  - Let $X = \begin{pmatrix} - & x_1 & - \\ \vdots & \vdots & \vdots \\ - & x_n & - \end{pmatrix}$, $Z = \begin{pmatrix} | & \dots & | \\ z_1 & \dots & z_k \\ | & \dots & | \end{pmatrix}$

# Linear Algebra

- Tool: We need an eigendecomposition from linear algebra.
  - Fact 1:
    - For any real symmetric nxn matrix A, one can write A as
      - $A=EDE^T$
    - Where D is a diagonal matrix with the i'th diagonal entry equal to the i'th largest eigenvalue of A, $\lambda_i$.
    - E is an orthonormal matrix, with i'th column = the unit-length eigenvector $\mu_i$ corresponding to $\lambda_i$. That is,
      - $A\mu_i = \lambda_i \mu_i$
    - All $\mu_i$ and $\lambda_i$ are real (not complex).
  - Fact 2:
    - If A is symmetric and positive semi-definite, then all eigenvalues are non-negative.

# PCA

- PCA: Given data $x_1,\dots,x_n$ choose $z_1,\dots,z_k$ to minimize
  - $\sum_{j=1}^{n} \left\| x_j - \sum_{i=1}^{k} (x_j^T z_i) z_i \right\|^2$

- Question:
  - How can we compute such $z_1,\dots,z_k$?
- Maximize:
  - $tr(Z^T X^T X Z)$

- Matrix Notation:
  - Let $X = \begin{pmatrix} - & x_1 & - \\ \vdots & \vdots & \vdots \\ - & x_n & - \end{pmatrix}$, $Z = \begin{pmatrix} | & \dots & | \\ z_1 & \dots & z_k \\ | & \dots & | \end{pmatrix}$

# PCA

- Examine $X^T X$:
  - X is nxd, so $X^T X$ is dxd.
- $$X^T X = \begin{pmatrix} x_{1,1} & \cdots & x_{n,1} \\ \vdots & \cdots & \vdots \\ x_{1,i} & \cdots & x_{n,i} \\ \vdots & \cdots & \vdots \\ x_{1,d} & \cdots & x_{n,d} \end{pmatrix} \begin{pmatrix} x_{1,1} & \cdots & x_{1,j} & \cdots & x_{1,d} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ x_{n,1} & \cdots & x_{n,j} & \cdots & x_{n,d} \end{pmatrix}$$
- Symmetric as entry (i,j) is inner product of i'th column in X with j'th column in X.
  - So entry (j,i) is inner product of j'th column in X with i'th column in X.

# PCA

- Examine $X^T X$:
  - X is nxd, so $X^T X$ is dxd.

- $$X^T X = \begin{pmatrix} x_{1,1} & \cdots & x_{n,1} \\ \vdots & \cdots & \vdots \\ x_{1,i} & \cdots & x_{n,i} \\ \vdots & \cdots & \vdots \\ x_{1,d} & \cdots & x_{n,d} \end{pmatrix} \begin{pmatrix} x_{1,1} & \cdots & x_{1,j} & \cdots & x_{1,d} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ x_{n,1} & \cdots & x_{n,j} & \cdots & x_{n,d} \end{pmatrix}$$

- Also positive semi-definite, i.e. for all z we have $z^T X^T X z \geq 0$:
  - $z^T X^T X z = (Xz)^T Xz = \|Xz\|_2^2 \geq 0$.

# Linear Algebra

- Tool: We need an eigendecomposition from linear algebra.
  - Fact 1:
    - For any real symmetric nxn matrix A, one can write A as
      - $A=EDE^T$
    - Where D is a diagonal matrix with the i'th diagonal entry equal to the i'th largest eigenvalue of A, $\lambda_i$.
    - E is an orthonormal matrix, with i'th column = the unit-length eigenvector $\mu_i$ corresponding to $\lambda_i$. That is,
      - $A\mu_i = \lambda_i \mu_i$
    - All $\mu_i$ and $\lambda_i$ are real (not complex).
  - Fact 2:
    - If A is symmetric and positive semi-definite, then all eigenvalues are non-negative.

# PCA

- Eigendecomposition of $X^T X$:
  - $X^T X = EDE^T$
- With:
  - $E = \begin{pmatrix} | & \dots & | \\ \mu_1 & \dots & \mu_d \\ | & \dots & | \end{pmatrix}$
  - $D = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \lambda_d \end{pmatrix}$
  - $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d \geq 0$.
  - $\|\mu_i\| = 1$ for all i.
  - $\mu_i^T \mu_j = 0$ for all $i \neq j$.

Goal: Choose $z_1,\dots,z_k$ as unit length and orthogonal to maximize

$$tr(Z^T X^T X Z)$$

With

$$Z = \begin{pmatrix} | & \dots & | \\ z_1 & \dots & z_k \\ | & \dots & | \end{pmatrix}$$

# PCA

- Eigendecomposition of $X^T X$:
  - $X^T X = EDE^T$
- With:
  - $E = \begin{pmatrix} | & \dots & | \\ \mu_1 & \dots & \mu_d \\ | & \dots & | \end{pmatrix}$
  - $D = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \lambda_d \end{pmatrix}$
  - $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d \geq 0$.
  - $\|\mu_i\| = 1$ for all i.
  - $\mu_i^T \mu_j = 0$ for all $i \neq j$.

Goal: Choose $z_1,\dots,z_k$ as unit length and orthogonal to maximize

$$tr(Z^T EDE^T Z)$$

With

$$Z = \begin{pmatrix} | & \dots & | \\ z_1 & \dots & z_k \\ | & \dots & | \end{pmatrix}$$

# PCA

- Eigendecomposition of $X^T X$:
  - $X^T X = EDE^T$
- With:
  - $E = \begin{pmatrix} | & \dots & | \\ \mu_1 & \dots & \mu_d \\ | & \dots & | \end{pmatrix}$
  - $D = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \lambda_d \end{pmatrix}$
  - $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d \geq 0$.
  - $\|\mu_i\| = 1$ for all i.
  - $\mu_i^T \mu_j = 0$ for all $i \neq j$.

Goal: Choose $z_1,\dots,z_k$ as unit length and orthogonal to maximize

$$tr(Z^T EDE^T Z)$$

With

$$Z = \begin{pmatrix} | & \dots & | \\ z_1 & \dots & z_k \\ | & \dots & | \end{pmatrix}$$

Let $A = E^T Z$, then must maximize

$$tr(A^T DA)$$

A is dxk.

# PCA

- Examine $tr(A^T DA)$:
- $DA = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \lambda_d \end{pmatrix} \begin{pmatrix} a_{1,1} & \dots & a_{1,k} \\ \vdots & \ddots & \vdots \\ a_{d,1} & \dots & a_{d,k} \end{pmatrix}$
- Scales i'th row of A by $\lambda_i$
- $DA = \begin{pmatrix} \lambda_1 a_{1,1} & \dots & \lambda_1 a_{1,k} \\ \vdots & \ddots & \vdots \\ \lambda_d a_{d,1} & \dots & \lambda_d a_{d,k} \end{pmatrix}$
- $A^T DA = \begin{pmatrix} a_{1,1} & \dots & a_{d,1} \\ \vdots & \ddots & \vdots \\ a_{1,k} & \dots & a_{d,k} \end{pmatrix} \begin{pmatrix} \lambda_1 a_{1,1} & \dots & \lambda_1 a_{1,k} \\ \vdots & \ddots & \vdots \\ \lambda_d a_{d,1} & \dots & \lambda_d a_{d,k} \end{pmatrix}$

i'th diagonal entry equals $\sum_{j=1}^{d} a_{j,i} \lambda_j a_{j,i} = \sum_{j=1}^{d} \lambda_j a_{j,i}^2$.

$tr(A^T DA)$ is $\sum_{i=1}^{k} \sum_{j=1}^{d} \lambda_j a_{j,i}^2 = \sum_{i=1}^{d} \lambda_i \sum_{j=1}^{k} a_{i,j}^2$

# PCA

- Eigendecomposition of $X^T X$:
  - $X^T X = EDE^T$
- With:
  - $E = \begin{pmatrix} | & \dots & | \\ \mu_1 & \dots & \mu_d \\ | & \dots & | \end{pmatrix}$
  - $D = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \lambda_d \end{pmatrix}$
  - $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d \geq 0$.
  - $\|\mu_i\| = 1$ for all i.
  - $\mu_i^T \mu_j = 0$ for all $i \neq j$.

Goal: Choose $z_1,\dots,z_k$ as unit length and orthogonal to maximize

$$tr(Z^T EDE^T Z)$$

With

$$Z = \begin{pmatrix} | & \dots & | \\ z_1 & \dots & z_k \\ | & \dots & | \end{pmatrix}$$

Let $A = E^T Z$, then must maximize

$$tr(A^T DA)$$

Equivalently maximize

$$\sum_{i=1}^{d} \lambda_i \sum_{j=1}^{k} a_{i,j}^2$$

# PCA

- Eigendecomposition of $X^T X$:
  - $X^T X = EDE^T$
- With:
  - $E = \begin{pmatrix} | & \dots & | \\ \mu_1 & \dots & \mu_d \\ | & \dots & | \end{pmatrix}$
  - $D = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \lambda_d \end{pmatrix}$
  - $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d \geq 0$.
  - $\|\mu_i\| = 1$ for all i.
  - $\mu_i^T \mu_j = 0$ for all $i \neq j$.

Goal: Choose $z_1,\dots,z_k$ as unit length and orthogonal and let $A=E^T Z$, then must maximize

$$\sum_{i=1}^{d} \lambda_i \sum_{j=1}^{k} a_{i,j}^2$$

With

$$Z = \begin{pmatrix} | & \dots & | \\ z_1 & \dots & z_k \\ | & \dots & | \end{pmatrix}$$

# PCA

Choose $z_1,\ldots,z_k$ as unit length and orthogonal and let $A=E^TZ$, then must maximize $\sum_{i=1}^{d}\lambda_i\sum_{j=1}^{k}a_{i,j}^2$

With

- $E=\begin{pmatrix}| & \cdots & |\\ \mu_1 & \cdots & \mu_d\\ | & \cdots & |\end{pmatrix}$
- $Z=\begin{pmatrix}| & \cdots & |\\ z_1 & \cdots & z_k\\ | & \cdots & |\end{pmatrix}$
- $\lambda_1\geq\lambda_2\geq\cdots\geq\lambda_d\geq 0$.
- $\|\mu_i\|=1$ for all i.
- $\mu_i^T\mu_j=0$ for all $i\neq j$.

# PCA

Choose $z_1,\ldots,z_k$ as unit length and orthogonal and let $A=E^TZ$, then must maximize $\sum_{i=1}^{d}\lambda_i\sum_{j=1}^{k}a_{i,j}^2$

With

- $E=\begin{pmatrix}| & \ldots & |\\ \mu_1 & \ldots & \mu_d\\ | & \ldots & |\end{pmatrix}$
- $Z=\begin{pmatrix}| & \ldots & |\\ z_1 & \ldots & z_k\\ | & \ldots & |\end{pmatrix}$
- $\lambda_1\geq\lambda_2\geq\cdots\geq\lambda_d\geq 0$.
- $\|\mu_i\|=1$ for all i.
- $\mu_i^T\mu_j=0$ for all $i\neq j$.

Linear Algebra Fact 1:

E and $E^T$ are orthonormal, so only rotates columns in Z.

Implies all columns in A have norm 1, i.e. $\sum_{i=1}^{d}a_{i,j}^2=1$.

Linear Algebra Fact 2:

i'th row of A has entries $\mu_i^Tz_j$ for j=1,..,k. So norm of i'th row is length of projection of $\mu_i$ onto subspace spanned by $z_1,\ldots,z_k$

# PCA

Choose $z_1,\dots,z_k$ as unit length and orthogonal and let $A=E^TZ$, then must maximize $\sum_{i=1}^{d}\lambda_i\sum_{j=1}^{k}a_{i,j}^2$

With

- $E=\begin{pmatrix}| & \dots & |\\ \mu_1 & \dots & \mu_d\\ | & \dots & |\end{pmatrix}$
- $Z=\begin{pmatrix}| & \dots & |\\ z_1 & \dots & z_k\\ | & \dots & |\end{pmatrix}$
- $\lambda_1\geq\lambda_2\geq\cdots\geq\lambda_d\geq 0$.
- $\|\mu_i\|=1$ for all i.
- $\mu_i^T\mu_j=0$ for all $i\neq j$.

Linear Algebra Fact 1:

E and $E^T$ are orthonormal, so only rotates columns in Z.

Implies all columns in A have norm 1, i.e. $\sum_{i=1}^{d}a_{i,j}^2=1$.

Linear Algebra Fact 2:

Norm of i'th row of A is length of projection of $\mu_i$ onto subspace spanned by $z_1,\dots,z_k$.

So $\sum_{j=1}^{k}a_{i,j}^2\leq\|\mu_i\|^2=1$.

# PCA

Choose $z_1,\ldots,z_k$ as unit length and orthogonal and let $A=E^TZ$, then must maximize $\sum_{i=1}^{d}\lambda_i\sum_{j=1}^{k}a_{i,j}^2$

With

- $E=\begin{pmatrix}| & \cdots & |\\ \mu_1 & \cdots & \mu_d\\ | & \cdots & |\end{pmatrix}$
- $Z=\begin{pmatrix}| & \cdots & |\\ z_1 & \cdots & z_k\\ | & \cdots & |\end{pmatrix}$
- $\lambda_1\geq\lambda_2\geq\cdots\geq\lambda_d\geq 0$.
- $\|\mu_i\|=1$ for all i.
- $\mu_i^T\mu_j=0$ for all $i\neq j$.

Observations:

$\sum_{j=1}^{k}\sum_{i=1}^{d}a_{i,j}^2=k$.

For all i: $\sum_{j=1}^{k}a_{i,j}^2\leq 1$.

Observations:

Everything in row i is scaled by $\lambda_i$.

We maximize $\sum_{i=1}^{d}\lambda_i\sum_{j=1}^{k}a_{i,j}^2$ if $\sum_{j=1}^{k}a_{i,j}^2=1$ for first k rows.

# PCA

Choose $z_1,\dots,z_k$ as unit length and orthogonal and let $A=E^TZ$, then must maximize $\sum_{i=1}^{d}\lambda_i\sum_{j=1}^{k}a_{i,j}^2$

With

- $E=\begin{pmatrix}| & \dots & |\\ \mu_1 & \dots & \mu_d\\ | & \dots & |\end{pmatrix}$
- $Z=\begin{pmatrix}| & \dots & |\\ z_1 & \dots & z_k\\ | & \dots & |\end{pmatrix}$
- $\lambda_1\geq\lambda_2\geq\dots\geq\lambda_d\geq 0$.
- $\|\mu_i\|=1$ for all i.
- $\mu_i^T\mu_j=0$ for all $i\neq j$.

Observations:

Everything in row i is scaled by $\lambda_i$.

We maximize $\sum_{i=1}^{d}\lambda_i\sum_{j=1}^{k}a_{i,j}^2$ if $\sum_{j=1}^{k}a_{i,j}^2=1$ for first k rows.

Solution:

Choose $z_i=\mu_i$ then entry (i,j) of A is $\mu_i^T\mu_j$.

# PCA

Choose $z_1,\ldots,z_k$ as unit length and orthogonal and let $A=E^TZ$, then must maximize $\sum_{i=1}^{d}\lambda_i\sum_{j=1}^{k}a_{i,j}^2$

With

- $E=\begin{pmatrix}| & \cdots & |\\ \mu_1 & \cdots & \mu_d\\ | & \cdots & |\end{pmatrix}$
- $Z=\begin{pmatrix}| & \cdots & |\\ z_1 & \cdots & z_k\\ | & \cdots & |\end{pmatrix}$
- $\lambda_1\geq\lambda_2\geq\cdots\geq\lambda_d\geq 0$.
- $\|\mu_i\|=1$ for all i.
- $\mu_i^T\mu_j=0$ for all $i\neq j$.

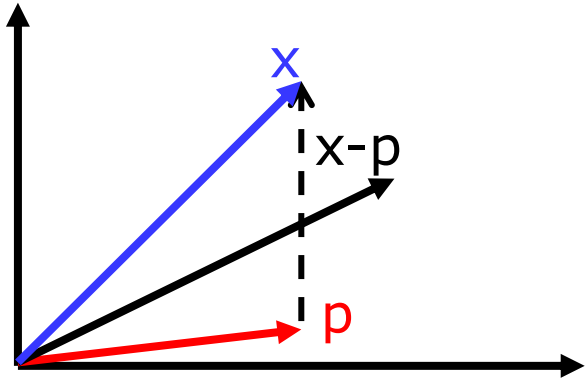
$$A=\begin{pmatrix}1 & \cdots & 0\\ \vdots & \ddots & \vdots\\ 0 & 0 & 1\\ \vdots & \ddots & \vdots\\ 0 & \cdots & 0\end{pmatrix}\Bigg\}\ \text{k}$$

# PCA

- Principal Component Analysis
  - Let $z_1,\ldots,z_k$ be k eigenvectors corresponding to largest eigenvalues of $X^TX$ and project feature vector x to $p_x = \sum_{i=1}^{k}(x^T z_i)z_i$.
- Observation:
  - This choice of $z_1,\ldots,z_k$ minimizes "squared loss" of projections:
    - $\sum_{j=1}^{n} \left\| x_j - p_{x_j} \right\|^2$
- Write x in the new basis:
  - $\hat{x} = \begin{pmatrix} x^T z_1 \\ \vdots \\ x^T z_k \end{pmatrix}$



MNIST



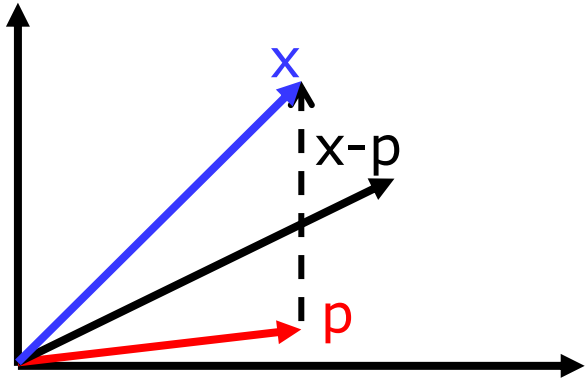
# Principal Component Analysis

# PCA

- Principal Component Analysis
  - Let $z_1,\ldots,z_k$ be k eigenvectors corresponding to largest eigenvalues of $X^TX$ and project feature vector x to $p_x = \sum_{i=1}^{k}(x^T z_i)z_i$.
- Observation:
  - This choice of $z_1,\ldots,z_k$ minimizes "squared loss" of projections:
    - $\sum_{j=1}^{n} \left\| x_j - p_{x_j} \right\|^2$
- Write x in the new basis:
  - $\hat{x} = \begin{pmatrix} x^T z_1 \\ \vdots \\ x^T z_k \end{pmatrix}$

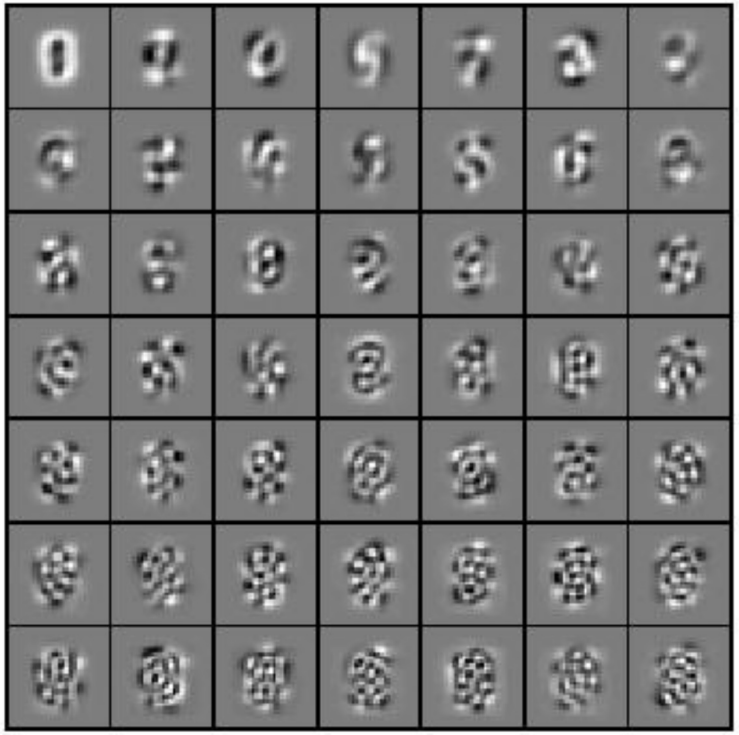


MNIST



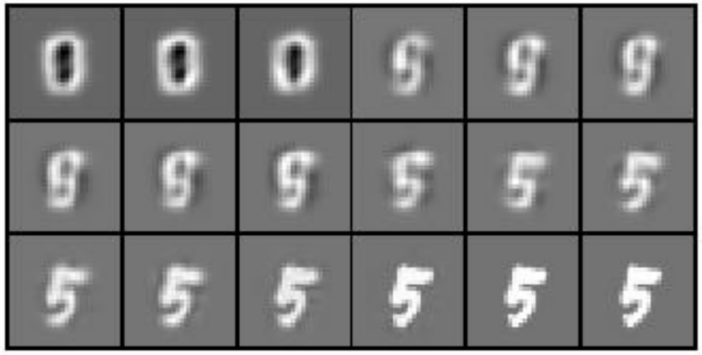
# PCA

PCA Basis (first 49)



5 Reconstruction with more and more basis vectors





Almost no loss with just a few basis vectors.

# Pitfalls

- Principal Component Analysis
  - Let $z_1,\ldots,z_k$ be k eigenvectors corresponding to largest eigenvalues of $X^TX$ and project feature vector x to $p_x = \sum_{i=1}^{k}(x^T z_i)z_i$.



- Careful!
  - Eigenvectors of matrix affected by scaling of matrix.

# Pitfalls

- Principal Component Analysis
  - Let $z_1,\ldots,z_k$ be k eigenvectors corresponding to largest eigenvalues of $X^TX$ and project feature vector x to $p_x = \sum_{i=1}^{k}(x^T z_i)z_i$.
  - Consider scaling j'th feature by a factor c:
  - $X = \begin{pmatrix} x_{1,1} & \cdots & cx_{1,j} & \cdots & x_{1,d} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ x_{n,1} & \cdots & cx_{n,j} & \cdots & x_{n,d} \end{pmatrix}$
  - j'th row in $X^TX$ scales by c, j'th column scales by c and j'th diagonal scales by $c^2$.
  - As c grows, diagonal (j,j) dominates whole matrix.

# Pitfalls

- Principal Component Analysis
  - Let $z_1,\ldots,z_k$ be k eigenvectors corresponding to largest eigenvalues of $X^TX$ and project feature vector x to $p_x = \sum_{i=1}^{k}(x^T z_i)z_i$.
  - Consider scaling j'th feature by a factor c:
  - $X^T X \approx c^2 \begin{pmatrix} 0 & \cdots & 0 \\ \vdots & 1 & \vdots \\ 0 & \cdots & 0 \end{pmatrix}$

Entry (j,j)

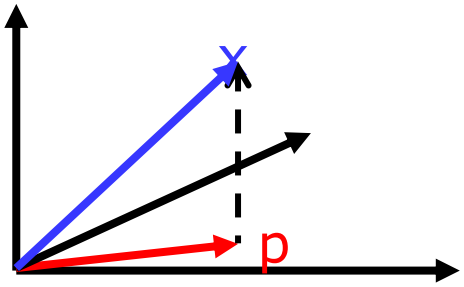
  - So $X^T X e_j \approx c^2 e_j$
  - That is, $z_1$ tends to $e_j$.
  - Effect: Will just keep j'th feature as it is.

# Pitfalls

- Principal Component Analysis
  - Natural Example with very different scaling:
  - Houses:
    - Price (4’200’000, 3’895’000,…
    - Bedrooms (3, 2,…
    - Square-meters (220, 165,…



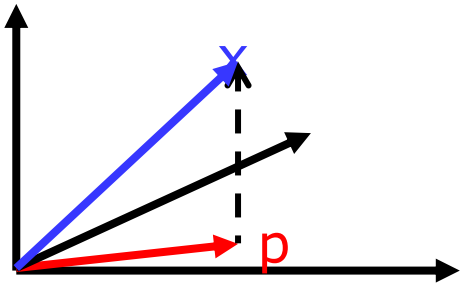
# PCA




- Principal Component Analysis
  - Start by scaling each feature such that mean $\mu = \sum_{i=1}^{n} \frac{x_{i,j}}{n}$ over data set is 0 and variance $\sum_{i=1}^{n} \frac{(x_{i,j}-\mu)^2}{n}$ is 1.

- Scaling a feature:
  - $\begin{pmatrix} x_{1,1} & \cdots & x_{1,j} & \cdots & x_{1,d} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ x_{n,1} & \cdots & x_{n,j} & \cdots & x_{n,d} \end{pmatrix}$
- To get mean 0:
  - Subtract $\mu_j = \sum_{i=1}^{n} \frac{x_{i,j}}{n}$ from each entry in j'th column.

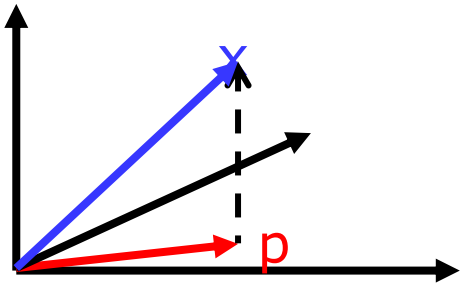
# PCA




- Principal Component Analysis
  - Start by scaling each feature such that mean $\mu = \sum_{i=1}^{n} \frac{x_{i,j}}{n}$ over data set is 0 and variance $\sum_{i=1}^{n} \frac{(x_{i,j}-\mu)^2}{n}$ is 1.

- Handling variance. Assume mean is already 0.
  - $\begin{pmatrix} x_{1,1} & \cdots & x_{1,j} & \cdots & x_{1,d} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ x_{n,1} & \cdots & x_{n,j} & \cdots & x_{n,d} \end{pmatrix}$
- Variance is: $\frac{1}{n}\sum_{i=1}^{n}(x_{i,j}-\mu)^2 = \frac{1}{n}\sum_{i=1}^{n} x_{i,j}^2$.
- Scale entries by $\sigma_j = \left(\frac{1}{n}\sum_{i=1}^{n} x_{i,j}^2\right)^{-1/2}$ then $\frac{1}{n}\sum_{i=1}^{n}(\sigma_j x_{i,j})^2$ becomes 1.

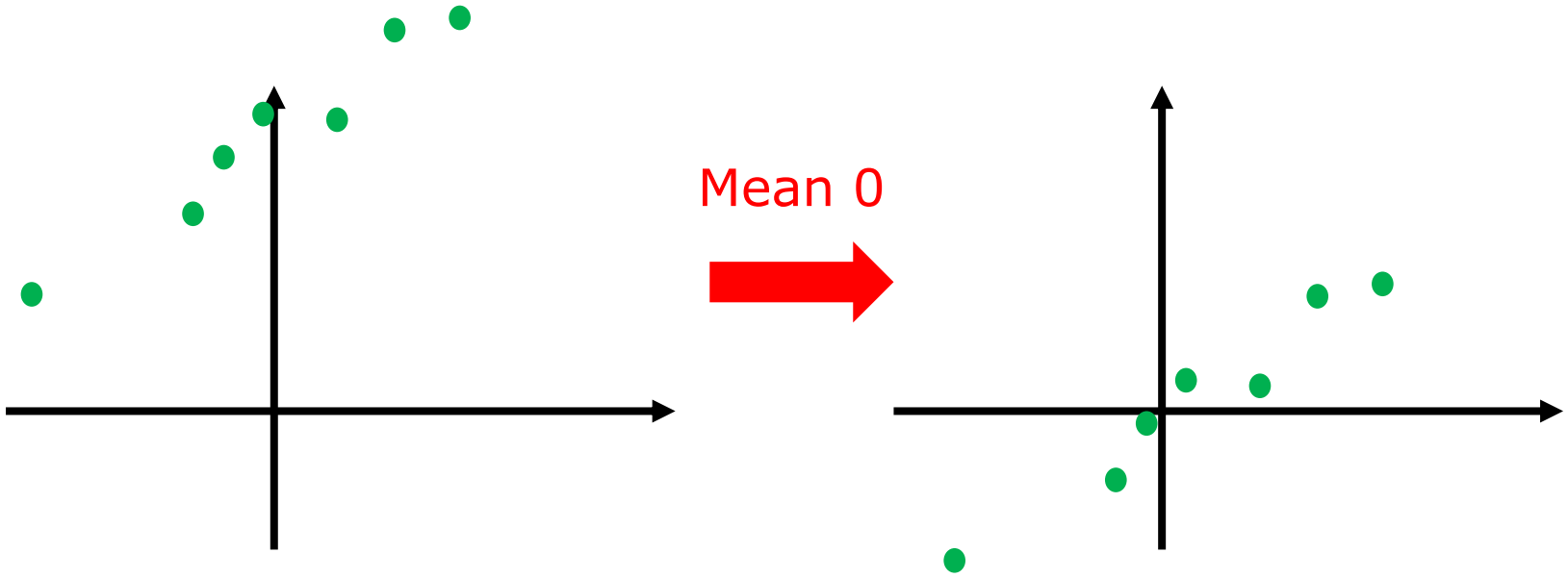
# PCA




- Principal Component Analysis
  - Subtract $\mu_j = \sum_{i=1}^{n} \frac{x_{i,j}}{n}$ from each entry in j'th column.
  - Scale entries by $\sigma_j = \left(\frac{1}{n}\sum_{i=1}^{n} x_{i,j}^2\right)^{-1/2}$ in j'th column.
  - Let $z_1,\ldots,z_k$ be k eigenvectors corresponding to largest eigenvalues of $X^TX$ and project feature vector x to $p_x = \sum_{i=1}^{k}(x^T z_i)z_i$.
  - Reduce dimensionality of input by replacing X by XZ.
- Run your favourite ML algorithm on new data XZ, e.g. Support Vector Machine, Linear Regression, Logistic Regression etc.
- To predict on new point x:
  - Subtract $\mu_j$ from j'th coordinate, scale j'th by $\sigma_j$ and compute $x^TZ$ (reduces to k features with correct scaling).

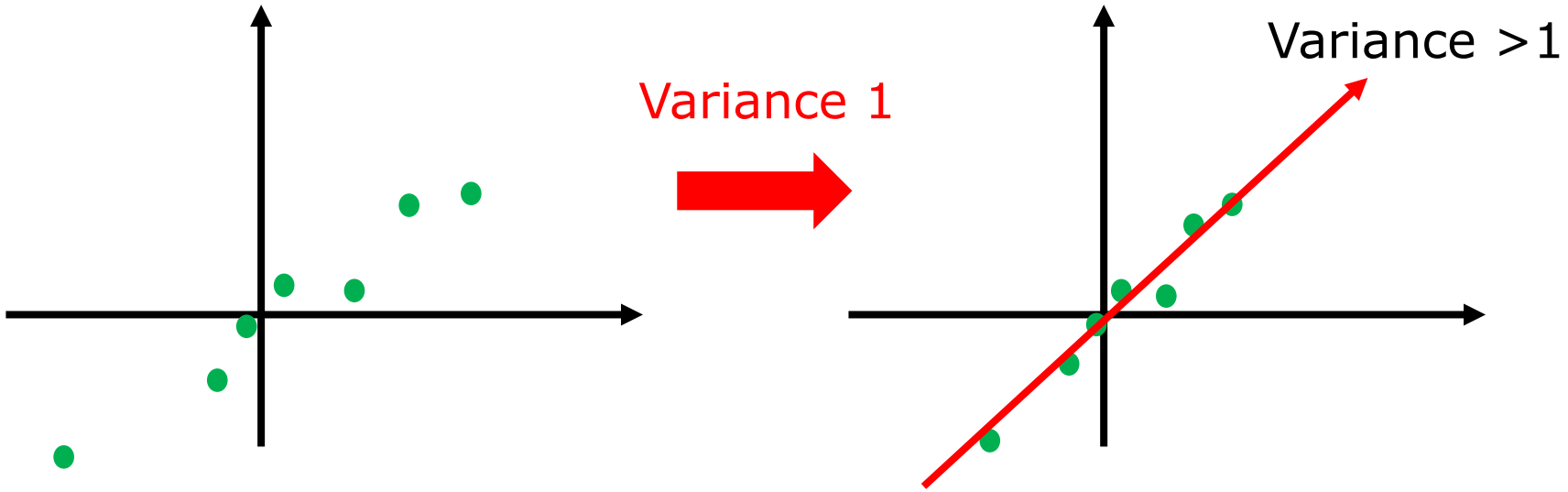
# Scaling

- Important Observation:
  - Only directions of standard unit vectors have variance 1.



# Scaling

- Important Observation:
  - Only directions of standard unit vectors have variance 1.




So still meaningful to choose directions of largest variance as basis.

# PCA

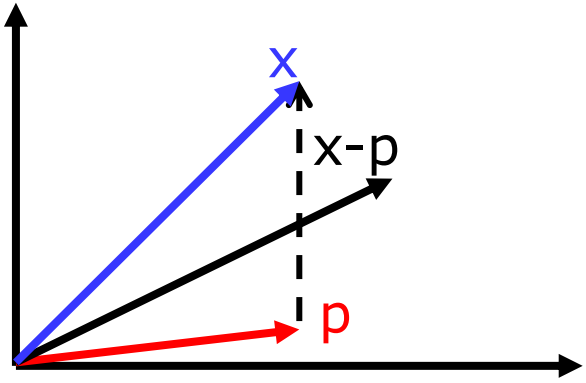
- Principal Component Analysis
  - Let $z_1,\ldots,z_k$ be k eigenvectors corresponding to largest eigenvalues of $X^TX$ and project feature vector x to $p_x = \sum_{i=1}^{k}(x^T z_i)z_i$.

- There are efficient algorithms for computing $z_1,\ldots,z_k$ ,for instance based on the Singular Value Decomposition of X.

- PCA is implemented in most learning software libraries.

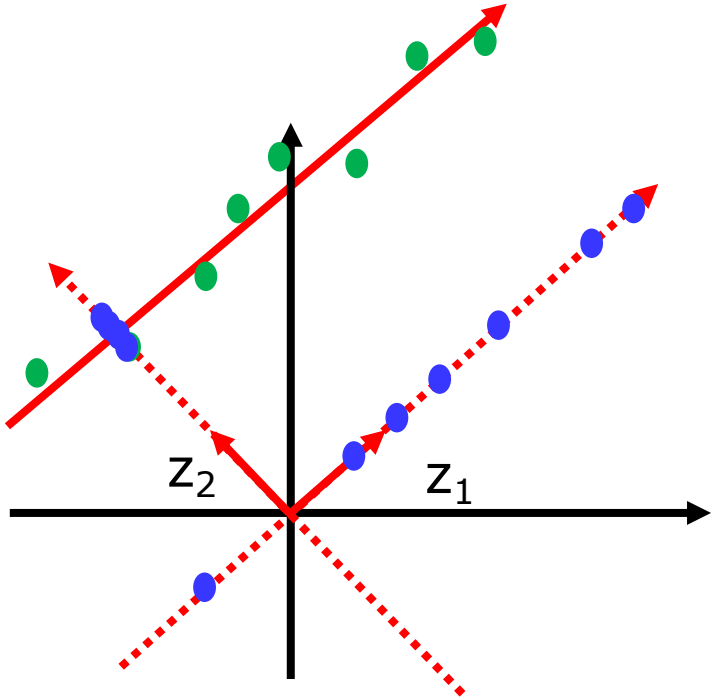


MNIST



# Understanding PCA

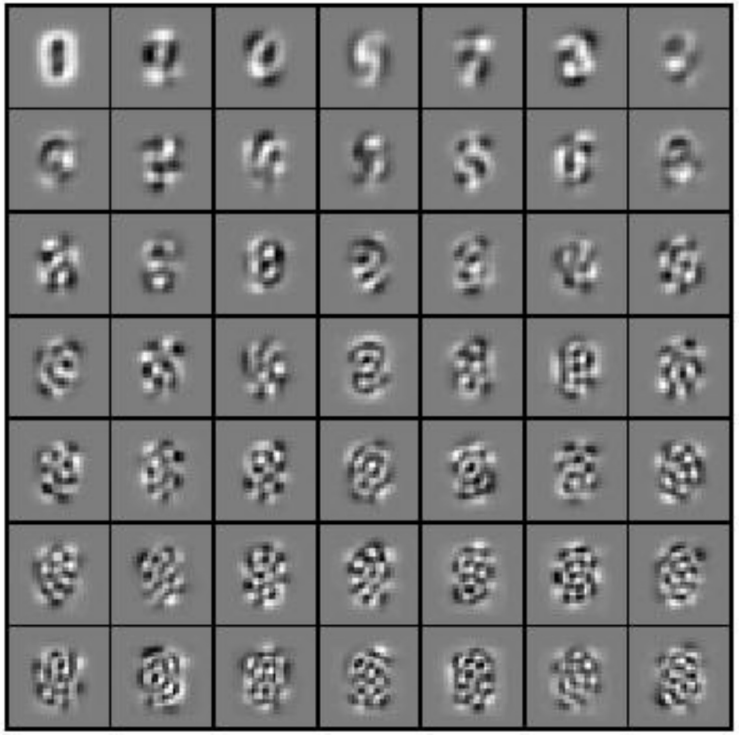
- Beyond Dimensionality Reduction
  - The vectors $z_1,\ldots,z_k$ were the directions of largest variance.




  - Often use PCA simply to obtain these directions as they may be used for understanding experimental data.

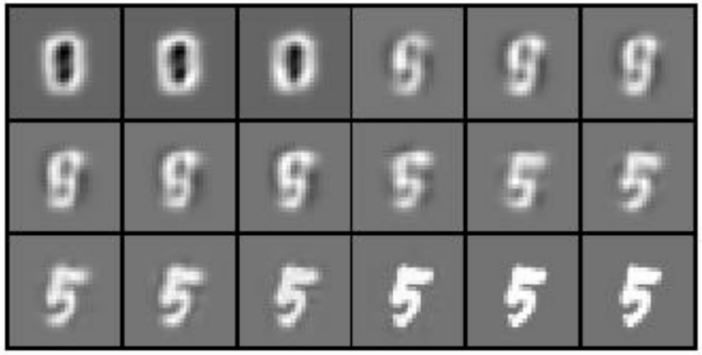
# PCA

PCA Basis (first 49)



5 Reconstruction with more and more basis vectors





Almost no loss with just a few basis vectors.